Age-appropriateness Classification of Czech Texts

Tomáš Novotný

ČVUT - FIT

novott37@fit.cvut.cz

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1 Introduction

This semestral project focuses on developing a model for assessing the age-appropriateness of texts in czech, with particular emphasis on content intended for children aged 8-13 in educational contexts. While existing research primarily addresses content inappropriateness detection, such as toxicity [1], [2] or erotic content [4], this project takes a more comprehensive approach by incorporating additional crucial factors such as cognitive load and language complexity. The goal is to create a sophisticated model that evaluates the multifaceted aspects of text appropriateness beyond mere content filtering.

2 Input Data

The training and evaluation data are sourced from the Czech National Corpus [3], which provides text samples categorized into two distinct classes,

- JUN: Literature specifically written for children
- GEN: Literature intended for the general public

The complete corpus comprises approximately 150,000 sentences labeled as JUN and 7.8 million sentences labeled as GEN. For experimental purposes, a balanced dataset containing 10,000 sentences from each category was created and is publicly available in the project repository. Data collection was performed through the KonText Interface using the following systematic pipeline:

1. Query: '.*' (retrieve all available data)
2. Constraint: 'audience:JUN/GEN' (filter by age-appropriateness label)
3. Filter: 'První nálezy ve větách' (eliminate duplicate entries)
4. Concordance: 'Náhodný vzorek' (generate random samples)
5. Display: 'KWIC/Věta' (sentence-level view)

The extracted data underwent preprocessing using a custom cleaning script to generate a normalized format with one sentence per line.

3 Methods

Given the project's focus on czech language classification, there was employed a monolingual contextualized language model [5] featuring 52,000 Czech-specialized tokens for sentence embeddings. This recently updated model demonstrates state-of-the-art performance on Czech language tasks. The classification component utilizes a dense MLP layer as its classification head.

As illustrated in Figure 1, the model's architecture encompasses approximately 125 million parameters. Following the standard RoBERTa architecture, the model incorporates 12 transformer layers, each comprising attention, intermediate, and output components. The model maintains a hidden dimension of 768 throughout most layers and employs the GELU activation function. To prevent overfitting, the model implements layer normalization after each major component and maintains a consistent dropout rate throughout its structure. The classification head employs a bottleneck architecture for final prediction.

3.1 Training

The implementation of both preprocessing and training pipelines is written in Python, primarily utilizing the transformers and torch libraries. The dataset was split into training (80%), validation (10%), and test (10%) sets. Table 1 details the hyperparameters used during training. To mitigate overfitting, several regularization techniques were implemented,

- dropout layers in both fully-connected and attention components,
- discriminative fine-tuning with lower learning rates in the model's lower layers,
- early stopping with patience of 2 epochs.

Table 1: Training parameters

parameter	value
max epochs	20
early stopping patience	2
1.-6. transformer layer learning rate	10^{-5}
7.-12. transformer layer learning rate	$3 \cdot 10^{-5}$
classifier head learning rate	$5 \cdot 10^{-5}$

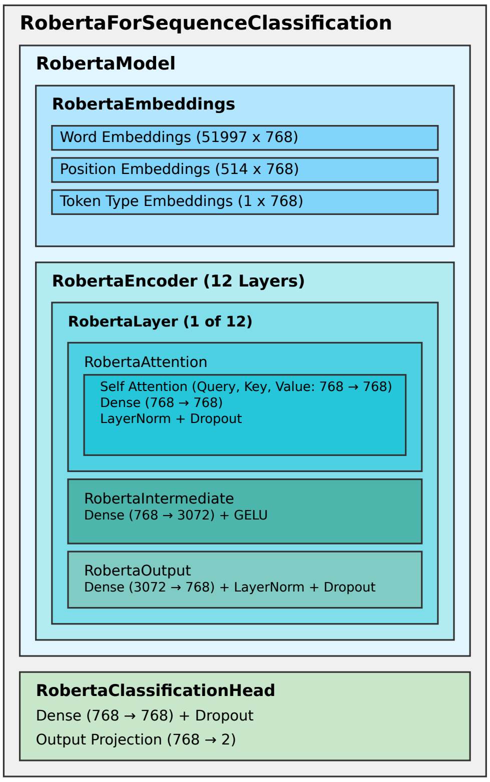

Figure 1: Architecture of the Classification Model

4 Results

The model achieved optimal performance after two epochs of training, as indicated by the minimum validation loss. The classification performance was robust across both classes, with Figure 2 illustrating the model's evaluation metrics on the test subset. The model achieved an accuracy of 76% and an F1-score of 0.78, demonstrating solid performance on this complex classification task.

Notably, the ROC AUC score of 0.84 is particularly significant, as it indicates the model's strong discriminative capability across different classification thresholds. This is especially relevant for age-appropriateness classification, where the optimal decision threshold may not necessarily be 0.50, but could be adjusted based on specific application requirements or risk tolerance levels.

To gain deeper insights into the model's behavior, detailed analysis is presented in the analysis notebook. This analysis focuses on investigating patterns in the most inaccurate misclassified examples and analyzing uncertain cases (predictions near 0.5).

5 Conclusion

The RobeCzech transformer architecture has demonstrated capability in distinguishing between children's and general public literature, validating its suitability for age-appropriateness classification. This model could serve as a component in a content filtering system, particularly when combined with specialized models for detecting erotic content and toxicity.

The relatively high performance achieved on this ambiguous task suggests promising potential for further development. However, several key areas for improvement and future work have been identified:

- **Dataset Enhancement:** Utilize the complete SYN15 dataset to create a more comprehensive and representative training corpus.
- **Data Quality:** Remove non-informative content (e.g., character names from dramatic works) and identify and properly handle overlap between children's and general public literature.
- **Interdisciplinary Validation:** Collaborate with linguists and pedagogical experts to evaluate the model's performance from linguistic and educational perspectives.

References

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[3] Michal Křen, Václav Cvrček, Tomáš Čapka, Anna Čermáková, Milena Hnátková, Lucie Chlumská, Dominika Kováříková, Tomáš Jelínek, Vladimír Petkevič, Pavel Procházka, Hana Skoumalová, Michal Škrabal, Petr Truneček, Pavel Vondřička, and Adrian Zasina. SYN2015: representative corpus of written

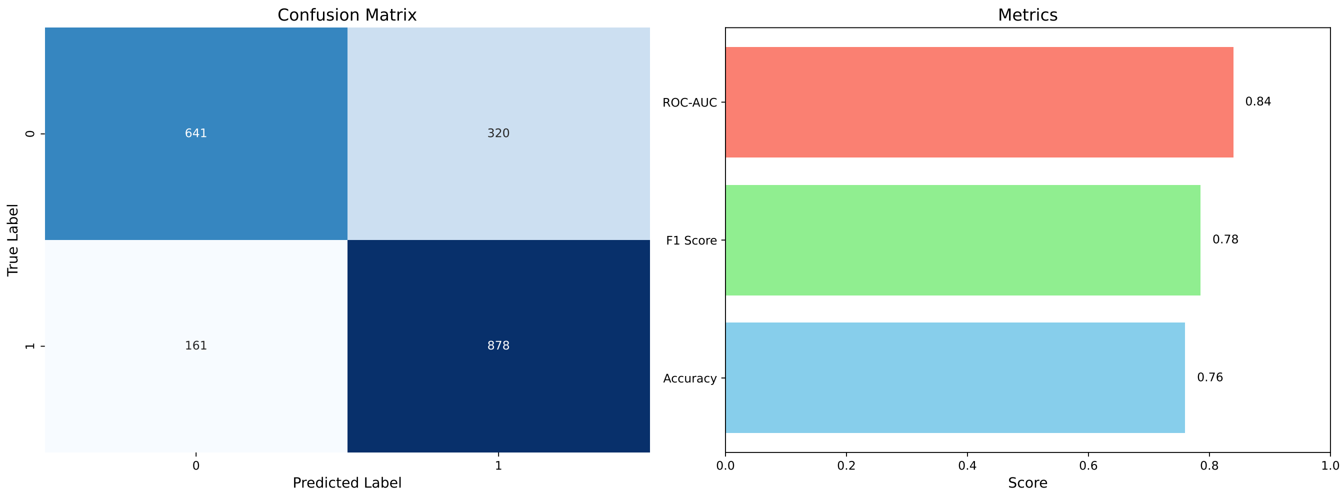

Figure 2: Evaluation of Model on Test Dataset

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